



School of Computer Science and Engineering

COURSE PLAN

Academic Year 2026-27 ODD SEMESTER

School/Department of Students	School of Computer Science and Engineering
Name of the Program(s) of Students	CAI-Computer Science & Engineering (Artificial Intelligence & Machine Learning) CCS-Computer Science & Engineering (Cyber Security) CIT-Computer Science & Engineering (Internet of Things) CSD-Computer Science & Engineering (Data Science) CBD-Computer Science & Technology (Big Data) CBC-Computer Science & Engineering (Block Chain) CDV-Computer Science & Technology (DevOps)
PRC Approval Ref. No.	PU/AC-24.7/SOCSE04/CSE/2023-2027
Semester/Year	V / III
Course Code & Name	CSS7000 & Internship
Credit Structure (L-T-P-C)	0-0-0-2
Contact Hours	NTCC
Course In-Charge (IC)	Dr. Geetha Arjunan
Course Instructor(s)	Internship Guides
Course URL	https://presidencyuniversity.linways.com

1. COURSE PRE-REQUISITES:

NIL

2. COURSE DESCRIPTION:

Students observe science and technology in action, develop an awareness of the method of scientific experimentation, and often get an opportunity to see, study and operate sophisticated and costly equipment. They also learn about the implementation of the principles of management they have learnt in class, when they observe multidisciplinary teams of experts from engineering, science, economics, operations research, and management deal with techno-economic problems at the micro and macro levels. Finally, it enables them to develop and refine their language, communication and inter-personal skills, both by its very nature, and by the various evaluation components, such as seminar, group discussion, project report preparation, etc. The broad-based core education, strong in mathematics and science and rich in analytical tools, provides the foundation necessary for the student to understand properly the nature of real-life problems.

3. COURSE OBJECTIVES:

The objective of the course is to familiarize the learners with the concepts of Professional Practice and attain Employability Skills through Experiential Learning techniques.

4. COURSE OUTCOMES:

TABLE 1: COURSE OUTCOMES		
CO Number	Statement of CO	Blooms Cognitive Level
	<i>On successful completion of the course the students shall be able to</i>	
CO1	Identify the engineering problems related to local, regional, national or global needs.	Understand
CO2	Apply appropriate techniques or modern tools for solving the intended problem	Apply
CO3	Design the experiments as per the standards and specifications.	Analyze
CO4	Interpret the events and results for meaningful conclusions.	Evaluate

5. MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

5.1 PROGRAM OUTCOMES:

On successful completion of the Program, the students will be able to:

- PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

- PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

TABLE 2a: CO-PO Mapping												
CO. No	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	M	M	M	L	L	L		M	H	M	L	M
CO2	H	M	H	M	H	M	L	M	H	M	L	M
CO3	H	L	H	M	H	L	L	L	H	M	M	H
CO4	M	H	M	H	M	L	L	L	H	M	M	H

5.2 PROGRAM SPECIFIC OUTCOMES:

On successful completion of the Program, the students will be able to:

PSO1	[Problem Analysis]: Identify, formulate, research literature, and analyze complex engineering problems related to Software Engineering principles and practices, Programming and Computing technologies reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
PSO2	[Design/development of Solutions]: Design solutions for complex engineering problems related to Software Engineering principles and practices, Programming and Computing technologies and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PSO3	[Modern Tool usage] : Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex

	engineering activities related to Software Engineering principles and practices, Programming and Computing technologies with an understanding of the limitations.
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TABLE 2b: CO-PSO Mapping			
CO Number	PSO1	PSO2	PSO3
CO1	M	L	L
CO2	H	H	M
CO3	M	H	H
CO4	M	M	H

6. ASSESSMENT SCHEDULE

TABLE 5: ASSESSMENT SCHEDULE							
Sl. No	Assessment Type	Coverage	CO Number(s)	Duration in Minutes	Marks	Weightage	Panel Members
1.	Final Viva (CA)	Submission of report & final presentation	CO1, CO2, CO3 & CO4	20 Minutes per student	100	100%	Project Guide & Domain Experts

7. COURSE CLEARANCE CRITERIA:

This is in accordance with the Academic Regulations of the University and the Program Regulations and Curriculum of the respective program.

8. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs): (Sample)

TABLE 7: SDG MAPPING			
S. No	Project Title	SDG Number	Justification
1	AI-Driven Crop Disease Prediction and Management System	SDG 2: Zero Hunger, SDG 9: Industry, Innovation and Infrastructure	SDG 2: Zero Hunger Justification: The system directly contributes to sustainable agriculture by improving crop yield and reducing crop loss due to diseases. Early disease prediction enables timely intervention, securing food supply and improving productivity — key aspects of ending hunger and ensuring food security. SDG 9: Industry, Innovation and Infrastructure Justification: Leveraging AI and machine learning for agricultural applications promotes innovation in agri-tech. It helps

			modernize traditional farming practices, fosters digital infrastructure in rural areas, and encourages research-driven solutions in the agricultural industry.
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9. CRITERIA FOR COURSE OUTCOME ATTAINMENT CALCULATION:

TABLE 8: Threshold and Target Set for Course Outcomes				
Sl. No	C.O. No.	Course Outcomes	Threshold in %	Target in %
1.	CO1	Identify the engineering problems related to local, regional, national or global needs.	70	74
2.	CO2	Apply appropriate techniques or modern tools for solving the intended problem	70	75
3.	CO3	Design the experiments as per the standards and specifications.	73	76
4.	CO4	Interpret the events and results for meaningful conclusions.	74	78

10. SUMMARY:

TABLE 9: SUMMARY OF COURSE SCHEDULE				
Sl. No.	Activity	Date	Time	Total number of Sessions
1.	Final Viva (CA)	22-8-2026	---	-

CONTACT TIMINGS IN THE CHAMBER FOR DISCUSSION

Students can meet with the respective Project Guide assigned during the Chamber Consultation Hour to clarify doubts related to the Internship.

SPECIFIC GUIDELINES TO STUDENTS, IF ANY:

- Students must select their project problem statements from the official list provided by the school, curated from SIH and MHRD sources. Each problem must be appropriately mapped to relevant Sustainable Development Goals (SDGs).
- Each project team shall consist of three students. No exceptions will be permitted.
- Students are required to immediately establish contact with their assigned guides upon notification of allocation.
- Attendance and participation in all scheduled reviews are mandatory. Each review carries marks as detailed in the approved review rubric.

- All reviews will be conducted offline (in-person).
- The review date and time must be coordinated with both the Project Guide and the assigned Reviewer. Each review must be conducted in the presence of the Reviewer within the allotted review window.
- A PowerPoint presentation (10–15 slides) is required for each review. This presentation will be assessed as part of the review.
- Students must begin preparing the project report from the outset of the semester. The draft should be reviewed periodically by the Guide.
- All students must adhere to the university's dress code and disciplinary norms during all reviews and interactions.
- As marks are allotted for research publication, students are expected to publish their project work in reputed journals or conferences to secure full marks in that component.
- Details regarding review panel members and venues will be communicated in due course.
- No makeup reviews will be allowed for any missed evaluation, including the Final Viva-Voce. Missing a review will result in loss of marks for that stage.
- After each review, all project teams must upload and organize their project artifacts—including source code, documentation, PPTs, datasets, etc.—in a GitHub repository. This will be reviewed by the Supervisor and Project Coordinator and will be a criterion for awarding review marks.

Evaluation Rubrics (Total: 100 Marks)

Evaluation Criteria	Description	Marks	Performance Indicators
Live Demonstration	Fully functional demo covering all features/modules.	50	Fully Functional (40–50) 75% Functional (30–39) Partially Working (20–29) Non-Functional (0–20)
Final Report (Hardcopy & Softcopy)	Professional report meeting PU formatting and guidelines.	10	As per Format (8–10) Partially Matched (4–7) Not Matched (0–3)
Mapping with Problem Statement based on company task	Ideas mapped to appropriate Problem and solutions	20	Mapped (20) Not Mapped (0)
GitHub Repository	Public GitHub repo with source code and documentation.	10	Excellent (9–10) Very Good (7–8.9) Good (5–6.9) Average (3–4.9) Below Average (0–2.9)
Answering Ability	Clarity and accuracy in answering technical questions.	10	Excellent (9–10) Very Good (7–8.9)

			Good (5–6.9) Average (3–4.9) Below Average (0–2.9)
	Total	100	

General Rating Scale Based on Weightage for Each Criterion

- Excellent (4): 85–100% of allocated marks
- Good (3): 70–84% of allocated marks
- Satisfactory (2): 50–69% of allocated marks
- Poor (1): Below less than 50% of allocated marks
- Absent/Not Reported (0): 0% of allocated marks



Dr Geetha Arjunan

Name and Signature of the course In-Charge

APPROVAL:

This course has been duly verified and approved by the Departmental Academic Committee (DAC).

Name and Signature of the Chairperson - DAC